ORIGINAL 5×3 SLOT MATHEMATICS · PORTFOLIO PAR SHEET · MODEL v1.1.0

Aether Loom

243 ways · physical 40-stop reel strips · persistent three-state Markov tension · finite 7–12 spin feature

Total RTP (exact)	Hit frequency (exact)	Feature frequency (exact)
96.5669%	**65.09%**	**1 in 237.74**

Volatility index, 90% (exact)	Max paid-cycle award (reachable)	10M-spin simulated RTP
6.35	**11,227.60×**	**96.6390%**

Item	Specification
Format	5 reels × 3 rows, 243 ways, left to right, 3+ consecutive reels
Bets	0.20, 0.50, 1.00, 2.00, 5.00, 10.00 fictional credits
Wild	Substitutes for all paying symbols; never on reel 1; no direct pay
Spark (scatter)	2 on a paid spin raise tension; 3+ trigger the feature; no direct pay
Tension	Rest / Taut / Overdrive; selects the next base reel set; stored per bet level, cleared on cash-out
Feature	7 free spins at the earned stage, cap 12; 2+ Sparks add a spin and raise the stage; multipliers 1.0× / 1.75× / 3.5×
Jackpot	None

Demonstration document for a fictional-credit portfolio game. Not a regulatory submission or certification. Configuration SHA-256 9959f5e5e14a. Every number is generated from reports/*.json by scripts/generate_docs.py.

1. Paytable (× total bet, per way)

Symbol	3 reels	4 reels	5 reels
CROWN	0.40	1.20	4.00
MOTH	0.30	0.90	2.50
SHUTTLE	0.20	0.60	1.50
DYE	0.15	0.40	1.00
LINEN	0.10	0.25	0.60
KNOT	0.07	0.15	0.40

2. RTP budget

Component	Exact
Base 243-ways wins	80.431642%
Feature	16.135273%
Total	**96.566915%**
Exact fraction, 20 dp	0.96566914833464226035

$$\mathrm{RTP} = \Sigma_s\, \pi_s\, (B_s + t_s\, F_s), \quad \pi P = \pi, \quad \Sigma \pi_s = 1$$

3. Base-state Markov chain

State	→ Rest	→ Taut	→ Overdrive	Stationary π	Conditional RTP	Value vs Rest
Rest	95.548%	4.452%	0.000%	93.2116%	91.29%	+0.000 bets
Taut	58.091%	34.091%	7.818%	6.2962%	153.09%	+1.184 bets
Overdrive	100.000%	0.000%	0.000%	0.4922%	372.09%	+2.755 bets

4. Feature by inherited stage

Start stage	Expected award	SD	Expected spins
Rest	32.4316×	35.990×	8.3705
Taut	61.4671×	45.986×	8.3705
Overdrive	96.0177×	50.032×	8.3705

5. Volatility and win distribution

Metric	Exact
Variance per paid spin	14.912377 bet^2
Standard deviation	3.861655× bet
Volatility index 90% / 95%	6.3519 / 7.5687
Long-run variance with state correlation $\sigma^2\infty$	14.831976 (-0.539%)
Hit frequency (any) / base	65.0918% / 64.8210%
Paid spins returning less than the bet	39.31%
Max base / max paid cycle (reachable)	88.00× / 11,227.60×

Base award (× bet)	Probability	1 in
0	35.1790%	2.8
(0, 1)	39.3069%	2.5
[1, 2)	13.7799%	7.3
[2, 5)	9.5891%	10.4
[5, 10)	1.9021%	52.6
[10, 20)	0.2331%	428.9
[20, 50)	0.0098%	10,191.4
[50, 100)	1.639e-08	61,008,686.8
100+	0 (impossible)	—

Exact single-spin distribution from all 40^5 stop combinations of each base reel set, weighted by π. Feature awards are not included in this table.

6. Player experience: RTP range after N spins

Paid spins	90% band	95% band
1,000	76.53% – 116.60%	72.70% – 120.44%
10,000	90.23% – 102.90%	89.02% – 104.12%
100,000	94.56% – 98.57%	94.18% – 98.95%
1,000,000	95.93% – 97.20%	95.81% – 97.32%
10,000,000	96.37% – 96.77%	96.33% – 96.81%

7. Verification

Check	Result
Float, rational, enumeration and independent Python RTP	agree; rational and Python fractions identical
Simulated RTP, 10,000,000 spins	96.6390% (z = +0.59, correlation-corrected)
95% CI half-width: i.i.d. / batch means / exact σ∞	±0.2402% / ±0.2582% / ±0.2387%
Chi-square, base win distribution	χ^2 = 5.64, df = 6, p = 0.464
Bet-switch player: shared state (v1.0) / per-bet state (v1.1)	153.47% / 96.74%

8. Reel strip symbol counts

Six named sets, five distinct strip configurations (BASE_OVERDRIVE = BONUS_REST). Ordered strips: docs/reel-strips.md.

BASE_REST

Symbol	R1	R2	R3	R4	R5
Crown	3	3	4	3	3
Moth	4	4	4	5	4
Shuttle	5	5	5	5	6
Dye	6	6	6	6	6
Linen	8	8	7	7	7
Knot	13	11	11	11	11
Wild	0	2	2	2	2
Spark	1	1	1	1	1

BASE_TAUT

Symbol	R1	R2	R3	R4	R5
Crown	3	3	4	3	3
Moth	4	4	4	5	4
Shuttle	5	5	5	5	6
Dye	6	6	6	6	6
Linen	8	7	7	6	7
Knot	13	10	10	10	10
Wild	0	3	3	3	3
Spark	1	2	1	2	1

BASE_OVERDRIVE

Symbol	R1	R2	R3	R4	R5
Crown	3	3	4	3	3
Moth	4	4	4	5	4
Shuttle	5	5	5	5	6
Dye	6	6	6	6	6
Linen	8	7	6	6	6
Knot	12	9	9	9	9
Wild	0	4	4	4	4
Spark	2	2	2	2	2

BONUS_REST

Symbol	R1	R2	R3	R4	R5
Crown	3	3	4	3	3
Moth	4	4	4	5	4
Shuttle	5	5	5	5	6
Dye	6	6	6	6	6
Linen	8	7	6	6	6
Knot	12	9	9	9	9
Wild	0	4	4	4	4
Spark	2	2	2	2	2

BONUS_TAUT

Symbol	R1	R2	R3	R4	R5
Crown	3	3	4	3	3
Moth	4	4	4	5	4
Shuttle	5	5	5	5	6
Dye	6	6	6	6	6
Linen	8	6	5	5	5
Knot	12	8	8	8	8
Wild	0	6	6	6	6
Spark	2	2	2	2	2

BONUS_OVERDRIVE

Symbol	R1	R2	R3	R4	R5
Crown	3	3	4	3	3
Moth	4	4	4	5	4
Shuttle	5	5	5	5	6
Dye	6	6	6	6	6
Linen	8	5	4	4	4
Knot	12	7	7	7	7
Wild	0	8	8	8	8
Spark	2	2	2	2	2